MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati
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Solutions - Tutorial Sheet No. 1 Date of Discussion: March 15, 2024

Lectures 1-4: Basics of ODEs, Formation of ODE, Separable ODE, Exact ODE, Integrating Factors, Special Nonlinear ODEs.

1. Determine the *order* and *degree* of the following ordinary differential equations. Also, identify these ODEs as *linear* or *nonlinear*.

   (i) $\dfrac{d^2y}{dx^2} + x\sin y = 0$

   (ii) $\dfrac{d^2y}{dx^2} + y\sin x = 0$

   (iii) $\left(1 + \dfrac{dy}{dx}\right)^{\frac{1}{2}} = x\dfrac{d^2y}{dx^2}$

   (iv) $\dfrac{d^6y}{dx^6} + \left(\dfrac{d^4y}{dx^4}\right)\left(\dfrac{d^3y}{dx^3}\right) + y = x$

   (v) $\sin\left(\dfrac{d^3y}{dx^3}\right) = x + y.$

   (vi) $\dfrac{d^3y}{dx^3} = \sin(x+y).$

   (vii) $\dfrac{d^3y}{dx^3} + \sin\left(\dfrac{d^2y}{dx^2}\right) = x + y$

   (viii) $\left(\dfrac{d^2y}{dx^2}\right)^2 + 4\dfrac{\left(\frac{d^2y}{dx^2}\right)^3}{\left(\frac{d^4y}{dx^4}\right)^2} + \dfrac{d^4y}{dx^4} = 0.$

   Solution:
   (i) Order 2, Degree 1, Nonlinear, (ii) Order 2, Degree 1, Linear, (iii) Order 2, Degree 2, Nonlinear, (iv) Order 6, Degree 1, Nonlinear, (v) Order 3, Degree not defined, Nonlinear, (vi) Order 3, Degree 1, Nonlinear, (vii) Order 3, Degree 1, Nonlinear, (viii) Order 4, Degree 3, Nonlinear.

2. Eliminating the arbitrary constants c_1 and c_2, obtain the differential equation satisfied by the following functions.
   (a) $y = c_1e^{-x} + c_2e^{2x}$
   (b) $x^2 + c_1y^2 = 1$
   (c) $y = c_1x - c_1^3$
   Solution:
   (a) Differentiating $y = c_1e^{-x} + c_2e^{2x}$ twice in succession,

$$y' = -c_1e^{-x} + 2c_2e^{2x},$$
$$y'' = c_1e^{-x} + 4c_2e^{2x}.$$

Eliminating c_1 and c_2 between them , we get

$$y'' - y' - 2y = 0.$$

(b) Differentiate $x^2 + c_1 y^2 = 1$ to get $x + c_1 y y' = 0$ and by taking $c_1 = (1 - x^2)/y^2$, the differential equation is obtained as

$$y' = \frac{xy}{x^2 - 1}.$$

(c) Differentiating $y = c_1 x - c_1^3$, we get $y' = c_1$. Eliminating c_1 from the given relation, we get the ODE as

$$y = xy' - (y')^3.$$

3. Are the following differential equations exact? If so, obtain their general solutions.

   (i) $(2xy - \sec^2 x)dx + (x^2 + 2y)dy = 0$

   (ii) $(x - 2xy + e^y)dx + (y - x^2 + xe^y)dy = 0$

   Solution:
   (i) Here $M(x, y) = 2xy - \sec^2 x$ and $N(x, y) = x^2 + 2y$. Clearly

   $$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}.$$

   Hence, the equation is exact and the solution can be obtained as

   $$\int (2xy - \sec^2 x)dx + \int 2y dy = c$$
   $$\Rightarrow x^2 y - \tan x + y^2 = c.$$

   (ii) Here $M(x, y) = x - 2xy + e^y$ and $N(x, y) = y - x^2 + xe^y$. Clearly

   $$\frac{\partial M}{\partial y} = -2x + e^y = \frac{\partial N}{\partial x}.$$

   Hence, the equation is exact and the solution can be obtained as

   $$\int (x - 2xy + e^y)dx + \int y dy = c$$
   $$\Rightarrow x^2/2 - x^2 y + xe^y + y^2/2 = c.$$

4. Solve the following ODEs:

   (i) $2x\, dx + 3(1 - y^3)\, dy = 0$

   (ii) $2xy\, dx + (x^2 + y^2)\, dy = 0$

   (iii) $x \sin y\, dx + (x + 1) \cos y\, dy = 0$

   (iv) $(y^2 + xy + x^2)\, dx - x^2\, dy = 0$

   (v) $\dfrac{dy}{dx} - 2(\cos x)y = \cos x,\ y(0) = \frac{1}{2}$

   (vi) $\dfrac{dy}{dx} + 2y = e^x$

Solution:
(i) It is a variable separable equation. The solution can be obtained as

$$x^2 + 3(y - 3y^4/4) = c.$$

(ii) It is an exact equation. The solution can be obtained as

$$\int 2xy dx + \int y^2 dy = c$$
$$\Rightarrow x^2y + y^3/3 = c.$$

(iii) It is variable separable. It can be written as

$$\left(1 - \frac{1}{x+1}\right) dx + \frac{\cos y}{\sin y} dy = 0$$

so as to get the solution as

$$x - \ln|x+1| + \ln|\sin y| = c.$$

(iv) It is a homogeneous differential equation. Putting $y = vx$, the differential equation becomes

$$\frac{dv}{v^2+1} = \frac{dx}{x},$$

which ultimately gives the solution as

$$\tan^{-1}\left(\frac{y}{x}\right) = \ln x + c.$$

(v) This is a linear equation whose integrating factor is found as $e^{-2\sin x}$. The solution is

$$y = -\frac{1}{2} + ce^{2\sin x}.$$

Upon using the given condition, $c = 1$ and hence, the solution is

$$y = -\frac{1}{2} + e^{2\sin x}.$$

(vi) This is a linear equation whose integrating factor is found as e^{2x}. The solution is

$$y = \frac{e^x}{3} + ce^{-2x}.$$

5. Find the values/conditions of the arbitrary constants/functions such that the following differential equations become exact, and hence solve them.

   (i) $(ax + by)dx + (cx + dy)dy = 0$

   (ii) $(x^2 + y^2)dx + (axy + y^4)dy = 0$

   (iii) $(2x + f(y))dx + 2xy\, dy = 0$

Solution:
(i) $b = c$, (ii) $a = 2$, (iii) $f(y) = y^2 + c, \quad c \in \mathbb{R}$.

6. Consider the following differential equation.

$$\frac{dy}{dx} = f(x)g(y),$$

where f is a continuous function with $f(x) \neq 0$ and g is a continuously differentiable function. Then, prove that $y(x) = c$ is a constant solution of the given ODE if and only if $g(y) = 0$.

Solution:

Part 1: Let $y \equiv c$ be a constant solution of the differential equation, that is, it solves $\dfrac{dy}{dx} = f(x, y)$. Now, $\dfrac{dy}{dx} = 0$ for $y \equiv c$. From the RHS of the differential equation, $f(x)g(c) = 0$ which implies that $g(c) = 0$ (since $f(x) \neq 0$).

Part 2: Conversely, if $g(c) = 0$, then consider the function $y(x) = c, \ \forall x$. Then, $y'(x) = 0$, that is, $f(x)g(y) = f(x)g(c) = 0$. This implies $y \equiv c$ satisfies the differential equation.

7. Show that the solution of $y' = k^2 y$, $y(x_0) = y_0$ is increasing if $y_0 > 0$ and decreasing if $y_0 < 0$.

Solution: The ODE can be easily solved to get $y = ce^{k^2 x}$. Using the given condition $y(x_0) = y_0$, it becomes $y = y_0 e^{k^2(x - x_0)}$. We know that the exponential function is always positive.

Hence, y is increasing if $y_0 > 0$, and
y is decreasing if $y_0 < 0$.

8. Consider the ODE $y' + ky = 1$, where k is a constant.
(a) For what value of k will all solutions tend to 2 as $x \to +\infty$?
(b) Is there any value of k for which there exists a non-constant solution $y(x)$ such that $y(x) \to -3$ as $x \to \infty$? Explain.

Solution: The solutioncan be obtained as

$$y = \frac{1}{k} + ce^{-kx}.$$

(a) $k = 1/2$ gives all solutions tending to 2 as $x \to +\infty$.
(b) There exists no value of k for which there exists a non-constant solution $y(x)$ such that $y(x) \to -3$ as $x \to +\infty$.

9. Solve the following Bernoulli's differential equations:

(i) $x\dfrac{dy}{dx} - y = x^k y^n, \quad n \neq 1, \ k + n \neq 1$

(ii) $2\dfrac{dy}{dx} + y = (x - 1)y^3$

(iii) $x^3\dfrac{dy}{dx} - x^2 y + y^4 \cos x = 0$

Solution:

(i) Substitute $z = y^{1-n}$. This gives a linear equation in z in the following form:

$$\frac{dz}{dx} - \frac{1-n}{x} z = (1-n)x^{k-1}.$$

The solution can be obtained as

$$z = (1-n)\frac{x^k}{k+n-1} + cx^{1-n}, \quad n \neq 1, k+n \neq 1.$$

It can be written in terms of y as

$$y^{1-n} = (1-n)\frac{x^k}{k+n-1} + cx^{1-n}, \quad n \neq 1, k+n \neq 1.$$

(ii) Substitute $z = y^{-2}$. This gives a linear equation in z in the following form:

$$\frac{dz}{dx} - z = 1 - x.$$

The solution can be obtained as

$$z = x + ce^x$$

It can be written in terms of y as

$$y^2 = \frac{1}{x + ce^x}.$$

(iii) Substitute $z = y^{-3}$. This gives a linear equation in z in the following form:

$$\frac{dz}{dx} + \frac{3}{x}z = \frac{3\cos x}{x^3}.$$

The solution can be obtained as

$$z = \frac{3\sin x + c}{x^3}.$$

It can be written in terms of y as

$$y^3 = \frac{x^3}{3\sin x + c}.$$

10. Write the following differential equations in Clairaut's form:

    (i) $\dfrac{dy}{dx} = \ln\left(x\dfrac{dy}{dx} - y\right)$

    (ii) $\dfrac{dy}{dx} = \cos^{-1}\left(y - x\dfrac{dy}{dx}\right)$

    Solution:
    (i)$y = xy' - e^{y'}$.
    (ii) $y = xy' + \cos y'$.

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